Lesson 3 Make a skeleton plot from measurements

About eight minutes, replacing the synthetic Exercise A on slides 14–16. These are the innermost 15 measured rings of well-frame boards A1 (29ho018) and C5 (29ho017), published by Nabunken. Equal ring indices do not imply equal calendar years.

Ring index	29ho018 (0.01 mm)	29ho017 (0.01 mm)
1	32	67
2	44	85
3	42	69
4	52	74
5	47	83
6	39	124
7	23	98
8	65	147
9	75	128
10	52	114
11	42	97
12	49	90
13	44	75
14	34	30
15	34	63

Mark conspicuously narrow rings in 29ho018. Draw equally spaced indices 1–15 and longer vertical marks for relatively narrower rings. Repeat for 29ho017 and compare the spacing between features.

Do not use measured width directly as mark length. Omit ordinary rings. This 15-ring excerpt teaches the procedure; it is not sufficient evidence for dating.

Measured data: Nabunken 29ho018, CC BY 4.0

Comparison specimen: 29ho017, CC BY 4.0

Lesson 3 Why withhold a date?

Example interpretation

In 29ho018, ring 7 (0.23 mm) is conspicuously narrow. Rings 14 and 15 (0.34 mm each) are narrower than ring 13 (0.44 mm), but the excerpt does not show the outer neighbour of ring 15. In 29ho017, ring 14 (0.30 mm) stands out. Mark lengths reflect relative judgements, so there is no unique numerical answer.

A partial match in a short excerpt does not establish a calendar date. Examine the full series, images and a securely dated reference. The records describe separate boards, but do not establish that they came from separate trees. Separate files are not necessarily independent biological replicates.

Read the Yoshino specimen coverage (slide 25)

Calendar year	Specimens present	Measured radii
1636	1	1
1650	2	3
1700	6	7
1800	24	30
2000	57	69
2018	13	18

How does replication differ between 1636 and 1800? Should two radii from the same tree count as two independent trees?

Counts are recalculated from the first and last years of specimens associated with the published study. Coverage describes replication, not a direct measure of dating correctness or reconstruction skill.

Yonenobu et al. (2026): Yoshino cedar